

National University of Computer and Emerging Sciences
Lahore Campus

Course: Linear Algebra

Date: 19-11-25

Course Instructor
Abdul Hafiz Sheikh

Quiz: 12

Total Time (Minutes): 2

Total Marks: 8

Name: _____ Roll No: _____ Section: _____

- The image of $x = (1, 1)$ under a rotation of $\pi/6$ about the origin is

$$R_{\pi/6} x = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}-1}{2} \\ \frac{1+\sqrt{3}}{2} \end{bmatrix}$$

- If the nullity of A is 4, then the nullity of $P^T A P =$ 4.

- If U and V are matrices, then $\langle U, V \rangle = \frac{U^T V}{U \cdot V}$.

- If $U = \begin{bmatrix} 5 & -1 \\ 2 & -2 \end{bmatrix}$ and $V = \begin{bmatrix} 1 & 3 \\ -1 & 0 \end{bmatrix}$, then

U and V are orthogonal

- Yes ✓ No • may or may not be

Good Luck!